



# VCS VERIFICATION REPORT

## QINGDAO LONGXIN WIND POWER PROJECT PHASE I

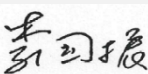


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China Quality Certification Centre

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## Summary:

China Quality Certification Centre(CQC) has performed the verification of the emission reductions reported for the project activity “VCS Verification Report of Qingdao Longxin Wind Power Project Phase I” (VCS Project ID:969) for the monitoring period 28/09/2012-20/12/2017, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity.

These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.3. The verification was performed on the basis of VCS Standard Version 4.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified 2CARs and no FAR is raised in this monitoring period. In CQC’s opinion, the GHG emission reductions reported for the project in the monitoring report(version 02 dated 30/08/2022) are fairly stated.

The Project has been registered as a CDM project on 28/09/2012 (UNFCCC registration reference number: 6697). The emission reductions of the project have not been issued as CERs.

CQC does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CQC and the engagement conditions detailed in this report.

CQC cannot be held liable by any party for decisions made or not made based on this report. Hence, CQC is able to certify that the emission reductions from the “Qingdao Longxin Wind Power Project Phase I” during this monitoring period from 28/09/2012- 20/12/2017 amount to 472,697 tCO<sub>2</sub>e.

# 1 INTRODUCTION

## 1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the “Qingdao Longxin Wind Power Project Phase I” for the period 28/09/2012- 20/12/2017.

## 1.2 Scope and Criteria

The scope of the verification is:

To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan;

To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;

To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

VCS Program Guide (version 4.2)<sup>/39/</sup>

VCS Standard (version 4.3)<sup>/38/</sup> and other relevant requirements defined by Verra;

The approved methodology ACM0002 (version 12.2.0)<sup>/42/</sup> applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate

## 1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.3, the project size categorization belongs to ‘Project’ as the

expected annual emission reductions is 92,183 tCO<sub>2</sub>e lower than 300,000 tCO<sub>2</sub>e. Therefore, according to the requirement of para 4.1.8 (4) of VCS Standard Version 4.3 “The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.”, CQC applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

## 1.4 Summary Description of the Project

Qingdao Longxin Wind Power Project Phase I (hereinafter referred to as the Project) developed by China Resources Wind Power (Qingdao) Co., Ltd., involves construction and operation of a wind power project that is sited on Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The Project installs 24 wind turbines with capacity of 2000kW each and 1 wind turbine with capacity of 1800kW, providing a total installed capacity of 49.8MW to generate clean renewable electricity.

The electricity generated by the Project is supplied to the North China Power Grid (NCPG), displacing electricity generated by NCPG which is dominated by fossil fuel fired power plants. The estimated annual net electricity generation and GHG emission reductions in the first crediting period (from 08/11/2011 to 07/11/2021) are 99,026MWh and 92,183tCO<sub>2</sub>e respectively.

The scenario existing prior to the start of the implementation of the project is: The same electricity output by the project activity would have otherwise been generated by the operation of NCPG connected power plants and by the addition of new generation sources. That is the same as the baseline scenario.

The construction of the Project started on 26/05/2011 and The Project was put into operation on 08/11/2011.

The project has been registered as VCS project with Ref. VCS 969 and VCUs of 82,302 tCO<sub>2</sub>e have been issued for the monitoring period from 08/11/2011- 27/09/2012. The Project has been registered as a CDM project on 28/09/2012 (UNFCCC registration reference number: 6697). The emission reductions of the project have not been issued as CERS.

During this monitoring period from 28/09/2012 to 20/12/2017, the total emission reduction achieved is 472,697tCO<sub>2</sub>e.

## 2 VERIFICATIONPROCESS

### 2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered VCS PD and monitoring plan contained in the registered VCS PD, the applied methodology, and the VCS Standard (version 4.3) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

### 2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CQC appointed a verification team in accordance with CQC's internal procedures.

| Function           | Name         | Technical competence | Task Performance*                                                                                                                                |
|--------------------|--------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Team Leader        | Zheng Xianyu | 1.1, 1.2, 2.1        | <input checked="" type="checkbox"/> DR <input checked="" type="checkbox"/> SV <input checked="" type="checkbox"/> RP <input type="checkbox"/> TR |
| Team Member        | Chi shuyang  | 1.1, 1.2             | <input checked="" type="checkbox"/> DR <input checked="" type="checkbox"/> SV <input checked="" type="checkbox"/> RP <input type="checkbox"/> TR |
|                    | Li Yanbin    | 1.1, 1.2             | <input checked="" type="checkbox"/> DR <input checked="" type="checkbox"/> SV <input checked="" type="checkbox"/> RP <input type="checkbox"/> TR |
| Technical Reviewer | Hong Dajian  | 1.2, 2.1, 13.1       | <input checked="" type="checkbox"/> DR <input type="checkbox"/> SV <input type="checkbox"/> RP <input checked="" type="checkbox"/> TR            |

\*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

During the desk review, CQC has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and an evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions

## 2.3 Interviews

On 09/08/2022, CQC visited the project proponent China Resources Wind Power (Qingdao) Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed and. Main topics of the interview are summarized in the table below:

| Interviewed personnel | Role               | Organization                                                        | Subject                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|--------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mr. Ju Bin            | Project Manager    | China Resources Wind Power (Qingdao) Co., Ltd.                      | Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement; Calibration of meters and equipment maintenance; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity. Project's sustainable development contributions; The impact of the project activity; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity; Project's sustainable development contributions. Data collection and ER calculation. |
| Mr. Shang Zitong      | Staff              | China Resources Wind Power (Qingdao) Co., Ltd.                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Mr. Pang Jianjian     | Staff              | China Resources Wind Power (Qingdao) Co., Ltd.                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Mr. Zhao Jie          | Staff              | China Resources Wind Power (Qingdao) Co., Ltd.                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Ms. Liu Yang          | Project Manager    | Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd. | Project's sustainable development contributions. Data collection and ER calculation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Ms. Ding Xue          | Technical director | Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## 2.4 Site Inspections

During the on-site assessment, CQC has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the registered VCS PD of the project activity;
- A review of information flows for generating, aggregating and reporting the monitoring parameters; and - Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity receipts;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the selected methodology;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 12.2.0), and the management system were assessed during the site visit.

## 2.5 Resolution of Findings

The objective of this phase of the verification is to resolve issues related to the monitoring, implementation and operations of the registered project activity that could impair the capacity of the project activity to achieve emission removals or influence the monitoring and reporting of emission removals prior to CQC's positive conclusion on the GHG emission removals calculation.

Findings established during the verification can either be seen as a non-fulfilment of criteria ensuring the proper implementation of a project or where a risk to deliver high quality emission reductions is identified.

A Corrective Action Request (CAR) is raised, if one of the following situations occurs:

- (a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;
- (b) Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project participants;
- (c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- (d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

A Clarification Request (CL) is raised, if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

A Forward Action Request (FAR) is raised, for actions if the monitoring and reporting require attention and/or adjustment for the next verification period.

To guarantee the transparency of the verification process, the concerns raised are documented in more detail in Appendix A.

2 Corrective Action Requests (CARs) had been raised during the verification, presented in Appendix A. Taking into account this output, the Project participant took corrections and revised its Monitoring Report (VR). All CARs are successfully closed. (refer to Appendix A for further details).

No FAR and no other findings raised during the validation.

### 2.5.1 Forward Action Requests



A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CQC confirmed that there was no FAR identified in previous verification<sup>/28/</sup>, and no FAR was raised during this verification.

## 2.6 Eligibility for Validation Activities

The project has been validated by TÜV Rheinland (China) Ltd. under the CDM rules and registered as CDM project on 28/09/2012, and was registered as VCS project under VCS standard version 3. The project has been registered as VCS project with Ref. VCS 969 and VCU of 82,302tCO<sub>2</sub>e have been issued for the monitoring period from 08/11/2011- 27/09/2012 by TÜV Rheinland (China) Ltd. The emission reductions of the project have not been issued as CERS.

CQC was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the proposed project. The monitoring period 28/9/2012-20/12/2017 is no more than consecutive 6 years. Thus, in compliance with the requirement for VVB stipulated in Section 4.1.20 of VCS standard version 4.3.

# 3 VALIDATION FINDINGS

## 3.1 Participation under Other GHG Programs

The project has been registered as CDM project and the reference number is 6697, the first CDM crediting period is from 28/09/2012 to 27/09/2019. The emission reductions of the project have not been issued as CERS.

The project has been registered as VCS project with Ref. VCS 969 and VCU of 82,302tCO<sub>2</sub>e have been issued for the monitoring period from 08/11/2011- 27/09/2012.

The project activity does not participated under other GHG programs.

## 3.2 Methodology Deviations

There is no methodology deviation applied during this monitoring period.

## 3.3 Project Description Deviations

In the registered VCS PD, the crediting period is described as from 08/11/2011 to 27/09/2012. A deviation is requested for the crediting period in the registered VCS PD. The project was registered under VCS Standard Version 3 and completed validation before 19/03/2020. Thus, it

remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice, so the crediting period of the project should be updated from 08/11/2011 - 27/09/2012 to 08/11/2011-07/11/2021, and it has no impacts on the applicability, additionality or baseline of the project. There is no other project description deviation during this monitoring period.

### 3.4 Grouped Project

The project is not a grouped project.

## 4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the “Qingdao Longxin Wind Power Project Phase I” for the monitoring period 28/09/2012-20/12/2017.

### 4.1 Project Implementation Status

#### **Project Implementation in accordance with the registered project design document**

The project is a wind power grid-connected plant and located at Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R.China. The geographical coordinates are verified by GPS system as from 119.8827°E to 119.9503°E, and from 36.8157°N to 37.8915°N (The central GPS coordinate is 119.91125°E, 36.8568°N).

This project installs 24 wind turbines with capacity of 2000 kW each and 1 wind turbines with capacity of 1800 kW which amount to a total installed capacity of 49.8MW. The estimated power output is 99,026MWh/year and the load factor is 22.70%. The Project started operation on 08/11/2011.

An 110kV substation is built at the project site, which connects the project to the Tangtian 220kV Substation which is about 6.7 km away from the project and belongs to the North China Power Grid. Then the power produced in the wind farm can be transmitted to the North China Power Grid.

Bidirectional meter M1 with accuracy 0.5S was installed on the Tangtian Substation to measure the total electricity supplied to the grid by the Project and Phase II project, meanwhile M1 is used to measure the total electricity imported from the grid by the Project and Phase II project. Meter M2 and M3 installed at the on-site substation recorded the electricity supplied by the Project and Phase II Project respectively.

Meter M4 is used to measure the electricity imported from the grid by the project and Phase II project via a spare 10kV back-up line.

During this monitoring period, the Project had been in normal operation, without special events such as overhaul or exchange of equipment. There were no events or situations which may impact the applicability of the methodology.

The monitoring system was implemented in line with the monitoring plan in the registered PD. The actual operation was conducted strictly in compliance with the description in the monitoring plan.

The actual implementation of the project during this verification period was verified in terms of nameplates of wind turbines and the equipment purchase contract. The details of the turbines with respect to their installation and capacity have been verified to be consistent with the description indicated in the registered VCS PD.

The electricity generated by the Project is supplied to the North China Power Grid (NCPG), displacing part of electricity generated by NCPG which is dominated by fossil fuel fired power plants. According to the methodology ACM0002 (version 12.2.0), the baseline emissions are the electricity produced by the project multiplied by the emission factor of NCPG.

All the monitoring system in operation period is consistent with the description in monitoring plan contained in the registered VCS PD. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the daily operation and maintenance records<sup>/7/</sup> and via on-site visit, CQC can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including, recording and reporting was verified to be in place<sup>/6/</sup> and their implementation was confirmed by interview with the key operators<sup>/43/</sup> and observing the operation.

CQC confirms that the project implementation is in accordance with the project description contained in registered VCS PD. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the monitoring plan.

### **Compliance of monitoring plan with monitoring methodology**

CQC is able to confirm that the monitoring plan contained in the VCS PD is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 12.2.0).

### **Compliance of monitoring with the monitoring plan**

The monitoring has been carried out in accordance with the monitoring plan contained in the VCS PD, CQC confirms that all parameters stated in the monitoring plan are monitored and reported appropriately. All parameters required to be monitored by the monitoring plan as per the monitoring methodology ACM0002 (version 12.2.0) and the management system were assessed during the site visit. The monitoring report lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following section.

#### **Parameters fixed ex ante**

“Data and parameters fixed ex ante” in the MR are checked against the registered VCS PD.

The parameter  $EF_{grid,CM,y}$  is the combined emission factor of the grid, which was determined ex-

ante at the validation stage and has been fixed during the first crediting period. CQC confirms that the emission factor used in the monitoring report is in compliance with the VCS PD.

### Parameters monitored

CQC conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and electricity receipts;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters

### According to the monitoring plan, the parameters to be monitored is as below

|                   |                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------|
| $EG_{output,y}$   | Electricity supplied to the grid by the Project and Phase II project                                       |
| $EG_{input,y}$    | Electricity consumed from the grid by the Project and Phase II project                                     |
| $EG_{another,y}$  | Electricity supplied to the grid by the Phase II project                                                   |
| $EG_{im-spare,y}$ | Electricity consumed by the Project and Phase II project from the grid the project through the backup line |

### Monitored parameters $EG_{output,y}$ and $EG_{input,y}$

By site inspection and checking the monitoring plan contained in the registered VCS PD, CQC confirmed  $EG_{output,y}$  and  $EG_{input,y}$  are continuously measured by meter M1. M1 is bi-directional and located at the high- voltage side of the on-site 35kV/110kV transformer station. The meter readings of electricity exports and imports were daily recorded, monthly aggregated and archived electronically.

The verification team had on-site checked the location of the meter against the diagram of power connection system <sup>/4/</sup> and found it to be consistent and installed in accordance with the approved monitoring plan. All the monitoring facilities and system have been verified by CQC during on-site visit.

By checking operation log <sup>/7/</sup> and via on-site visit, it is confirmed that M1 was working well during this monitoring period. Via the on-site interview and by checking the daily operation and maintenance records <sup>/7/</sup>, monthly and daily reading records and electricity receipts <sup>/9/</sup> issued by grid company, it is confirmed by the verification team that the electricity is continuously measured by M1, daily recorded and monthly aggregated by trained staff of the project as agreed with the grid

company at 24:00 on the aggregating, the last day of each month <sup>/8/</sup>. The electricity receipts<sup>/9/</sup> issued by the grid company have been used for verifying the records of  $EG_{output,y}$  and  $EG_{input,y}$  measured by main meter M1.

#### Monitored parameter $EG_{another,y}$

By site interview and checking the monitoring plan contained in the registered VCS PD and previous verification reports, CQC confirmed  $EG_{another,y}$  is continuously measured by meter M3, which is installed on the 110kv outline of the main transformer of Phase II project on site. The meter reading was daily recorded, monthly aggregated and archived electronically. By checking operation log <sup>/7/</sup> monthly and daily reading records <sup>/8/</sup> and via on-site visit, it is confirmed that M3 was working well during this monitoring period and the electricity is continuously measured by M3, daily recorded and monthly aggregated by trained staff of the project as agreed with the grid company at 24:00 on the aggregating, the last day of each month.

By site interview and checking the monitoring plan contained in the registered VCS PD and previous verification reports, CQC confirmed  $EG_{im-spare,y}$  is continuously measured by meter M4, which is installed on the 10kv backup line on site. The meter reading was daily recorded, monthly aggregated and archived electronically. By checking operation log <sup>/7/</sup> monthly and daily reading records <sup>/8/</sup> and via on-site visit, it is confirmed that M4 was working well during this monitoring period and the electricity is continuously measured by M4, daily recorded and monthly aggregated by trained staff of the project as agreed with the grid company at 24:00 on the aggregating, the last day of each month.

As described above, the meters have been installed in accordance with VCS Program rules and meet all appropriate rules and requirements of VCS standard. CQC has on-site checked the location of the meters against the diagram of power connection system and found them to be consistent.

Data in the monthly reading records were used to the report, through a cross check with Electricity Receipts. The conservative values were used for the monitoring report and ERs calculation spread sheet, which has been verified by the verification team. Supporting references and data required to determine the quantity of net electricity generation supplied by the project is found to be complete and transparent.

**CAR 01** was raised because of the monitoring period in the MR (from 28/09/2012 to 07/11/2021, which is more than six consecutive years) does not meet the requirements of rotation of validation/verification bodies in VCS-Standard\_v4.3. The monitoring period has revised from 28/09/2012-07/11/2021 to 28/09/2012-20/12/2017. **CAR 01** was subsequently successfully closed.

**CAR 02** was raised because of the information of the monitoring meters installed during the monitoring period is not provided in the MR and supporting documents. The information of the monitoring meters during this monitoring period have been added in the section 4.1 of the revised MR (V02, 30/08/2022). The main information are including the type, accuracy, serial number,

calibration date, validity period and name of the calibration organization, and the meters' photo and the calibration reports covering this monitoring period have been submitted. **CAR 02** was subsequently successfully closed.

Refer to Appendix B for details.

### **Monitoring equipment and calibration**

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the methodology ACM0002(version 12.2.0) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the monitoring plan covering this monitoring period.

The calibration frequency of meters is annual, which is consistent with the monitoring plan. Based on the site visit and by checking the calibration reports, CQC confirms that:

- All meters have been installed in accordance with the monitoring plan;
  - The meters' accuracy is in line with the requirement of the monitoring plan;
  - The calibration frequency of the meters is annual, which is in line with the monitoring plan.
- And the calibrations are verified to be valid for the whole reporting period.

### **Data management and control**

The quality assurance and quality control procedures have been addressed in the project management and monitoring manual<sup>/5/</sup>, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/43/ and check records<sup>/7//8/</sup> during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification.

Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and monitoring plan contained in the VCS PD and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction(CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification
- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification

The project is completely voluntary VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and energy intensive industries that emit at least 26,000 tons of CO<sub>2</sub>e/year as per the regulation<sup>/21/</sup> including thermal power generation, petrochemical, chemical, building materials, iron and steel, etc. By checking the business license<sup>/16/</sup> of the project proponent, it is confirmed in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no emission cap enforced for the PP by further checking the enforced company list under China ETS in public information<sup>/21/</sup>. Moreover, it is confirmed by CQC the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation<sup>/20/</sup>. Hence, it is confirmed that the emission reductions will not be double counted.

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- SDG 7, Indicator 7.2.1 Renewable energy share in the total final energy consumption.  
The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in the grid NCPG. During this monitoring period, 507,789.778 MWh of net electricity generated by wind energy was delivered to the NCPG. It is verified by checking the daily and monthly reading records<sup>/8/</sup> and via on-site inspection. Meanwhile, by checking the approved VCS and CDM monitoring reports for the previous monitoring, the project has supplied 596,207.338 MWh renewable energy to NCPG during the first monitoring period and this monitoring period. And 1,980,520 MWh electricity from renewable source are expected to supply to NCPG over the project lifetime which contribute the SDGs 7. This contributes to increasing the share of clean energy consumption of China's National Plan on Implementation of 2030 Agenda for Sustainable Development (<http://www.gov.cn/xinwen/2016-10/13/5118514/files/44cb945589874551a85d49841b568f18.pdf>).
- SDG 8, Indicator 8.3.1 Proportion of informal employment in non agriculture employment, by sex.  
It is verified by interviewing with project proponent and staff and checking the staff roster<sup>/22/</sup>, 22 people including 6 females were employed by the project owner for the operation,



monitoring and equipment maintenance of this project during this monitoring period. The project activity had provided temporary working opportunities during the construction period. From the operation start date, 22 people were employed including 6 females during the first VCS monitoring period and this monitoring period. This contributes to "Increase labor force participation rate through implementation rate through implementation of the classification policy. Vigorously enforce the law on promotion of employment" as stated in China's National Plan on Implementation of 2030 Agenda for Sustainable Development (<http://www.gov.cn/xinwen/2016-10/13/5118514/files/44cb945589874551a85d49841b568f18.pdf>).

- SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed. As verified by on-site inspection, and via checking the emission reduction calculation sheet<sup>/22/</sup> and the daily and monthly reading records<sup>/8/</sup>, 472,697tCO<sub>2</sub>e GHG emission reductions are generated by the project activity during this monitoring period. And by checking the VCS and CDM monitoring and verification reports for the previous monitoring periods<sup>/27/</sup>, it is confirmed the cumulative emission reductions are 554,999 tCO<sub>2</sub>e during the first VCS monitoring period and this monitoring period. And according to the registered PDD, the project was expected to prevent the release of 1,740,680 tCO<sub>2</sub>e into the atmosphere during the project operation lifetime of 20 years. This contributes to achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas".

In conclusion, Verification Team was able to confirm that the project implementation is in accordance with the project description contained in the VCS PD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

## 4.2 Safeguards

### 4.2.1 No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved VCS PD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and residents, wastewater, noise, solid waste and atmosphere etc. is very little, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent, staff and local villagers and checking the staff roster and monthly payroll<sup>/22/</sup>, the Explanation of the employment offered by the project<sup>/23/</sup> and national poverty standard<sup>/24/</sup>, that the project has contributed to the local employment and improved income levels by providing job opportunities and offering permanent employees wages well above poverty income levels. As stated by the plant staff during the on-site interviews, they



have gained knowledge and skills in the regular training organised by the project proponent, so it can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them.

#### 4.2.2 Local Stakeholder Consultation

The local stakeholder's meeting was held in Zhujiaying Village on Mar 2011. 49 participants attended the meeting including local residents, builders and members of the local authorities. The project owner introduced the proposed project, and then a survey was arranged through a one-page questionnaire, which was designed to be easily filled in. The opinions expressed by the stakeholders were recorded and are available on request.

The stakeholder meeting and the survey showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.

For continuous communication with local stakeholders, the project owner public its office telephone to local people. Through interviewing with staff from local environmental protection bureau and local residents during the site visit, it is confirmed by the verification team that during the implementation stage of this monitoring period, local authority has conducted spot checks on the implementation of the project periodically as per the request from the local governments' regulations, no negative comments and issues from the local stakeholders during this monitoring period were identified and the project passed all the periodic spot checks by local government.

All such conclusion has been verified through site visit and checking VCS PD.

#### 4.3 AFOLU-Specific Safeguard

The project is not AFOLU project.

#### 4.4 Accuracy of GHG Emission Reduction and Removal Calculations

CQC confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$$ER_y = BE_y - PE_y - L_y$$

##### **Baseline emissions**

Baseline emissions are determined as multiplying quantity of net electricity supplied to the grid by the project during year y ( $EG_y$ ) by the validated ex-ante fixed grid emission factor ( $EF_{grid,CM,y}$ ).

$$BE_y = EG_y \times EF_{\text{grid,CM,y}}$$

Grid emission factor ( $EF_{\text{grid,CM,y}}$ )

$EF_{\text{grid,CM,y}}$  is the grid emission factor of the which has been verified ex-ante in the validation stage in the VCS PD as 0.9309 tCO<sub>2</sub>e/MW

The net electricity supplied to the grid by the project ( $EG_y$ )

As per the Section 4.1 above,  $EG_y$  is determined as:

The verification team has checked the daily and monthly reading records, electricity receipts, Confirmation letter for electricity readings issued by the grid company and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The reading records and  $EG_y$  calculation are summarized in Table 1 as follow :

**Table 1 Net electricity supplied to the Grid by the project**

| Monitoring period    |            | Total electricity supplied to the Grid by the Project and Phase II Project (EG <sub>output,y</sub> ) unit: MWh |                             |                    | Total electricity imported from the Grid by the project and Phase II project through the 110kV line (EG <sub>import,y</sub> ) unit: MWh |                              |                     | Electricity imported from the Grid by the project and Phase II project through a spare 10 kV line (EG <sub>im-spare</sub> ) unit: MWh |                              |                     | Electricity generated by the Phase II Project (EG <sub>another,y</sub> ) unit: MWh | Net electricity supplied to the grid by the project (EG <sub>facility</sub> ) unit: MWh                         |
|----------------------|------------|----------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| From                 | To         | Data based on meter readings M1                                                                                | Data based on sales receipt | Conservative value | Data based on meter readings M1(110kV)                                                                                                  | Data based on sales receipts | Conservative values | Data based on meter readings M4(10kV)                                                                                                 | Data based on sales receipts | Conservative values | Data based on meter readings M3                                                    | Calculated by EG <sub>output,y</sub> -EG <sub>import,y</sub> -EG <sub>im-spare,y</sub> +EG <sub>another,y</sub> |
|                      |            | A                                                                                                              | B                           | C=Min (A, B)       | D                                                                                                                                       | E                            | F=Min (D, E)        | G                                                                                                                                     | H                            | I=Max (G, H)        | J                                                                                  | J=C-F-I-J                                                                                                       |
| 28/09/2012           | 25/10/2012 | 15800.810                                                                                                      | 15800.810                   | 15800.810          | 39.327                                                                                                                                  | 102.004                      | 102.004             | 0.429                                                                                                                                 | 0.429                        | 0.429               | 8682.080                                                                           | 7,016.297                                                                                                       |
| 26/10/2012           | 25/11/2012 | 21600.480                                                                                                      | 21600.480                   | 21600.480          | 30.360                                                                                                                                  | 29.04                        | 30.360              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 11698.720                                                                          | 9,870.939                                                                                                       |
| 26/11/2012           | 23/12/2012 | 21933.120                                                                                                      | 21933.120                   | 21933.120          | 27.720                                                                                                                                  | 35.640                       | 35.640              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 12515.360                                                                          | 9,381.659                                                                                                       |
| 24/12/2012           | 31/12/2012 | 4709.760                                                                                                       | 4709.760                    | 4709.760           | 7.920                                                                                                                                   | 6.93                         | 7.920               | 0.115                                                                                                                                 | 0.115                        | 0.115               | 2442.880                                                                           | 2,258.845                                                                                                       |
| <b>Subtotal 2012</b> |            |                                                                                                                |                             | <b>64044.170</b>   |                                                                                                                                         |                              | <b>175.924</b>      |                                                                                                                                       |                              | <b>1.466</b>        | <b>35339.040</b>                                                                   | <b>28,527.740</b>                                                                                               |
| 01/01/2013           | 25/01/2013 | 19315.560                                                                                                      | 19315.560                   | 19315.560          | 23.760                                                                                                                                  | 20.790                       | 23.760              | 0.346                                                                                                                                 | 0.346                        | 0.346               | 9368.480                                                                           | 9,922.974                                                                                                       |
| 26/01/2013           | 25/02/2013 | 11820.600                                                                                                      | 11820.600                   | 11820.600          | 151.800                                                                                                                                 | 149.160                      | 151.800             | 0.461                                                                                                                                 | 0.461                        | 0.461               | 7084.000                                                                           | 4,584.339                                                                                                       |
| 26/02/2013           | 26/03/2013 | 26550.480                                                                                                      | 26550.480                   | 26550.480          | 15.840                                                                                                                                  | 18.480                       | 18.480              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 14490.080                                                                          | 12,041.459                                                                                                      |
| 27/03/2013           | 24/04/2013 | 28005.120                                                                                                      | 28005.120                   | 28005.120          | 6.600                                                                                                                                   | 10.560                       | 10.560              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 14414.400                                                                          | 13,579.699                                                                                                      |
| 25/04/2013           | 25/05/2013 | 24845.040                                                                                                      | 24845.040                   | 24845.040          | 15.840                                                                                                                                  | 14.520                       | 15.840              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 13221.120                                                                          | 11,607.619                                                                                                      |
| 26/05/2013           | 24/06/2013 | 19561.080                                                                                                      | 19561.080                   | 19561.080          | 21.120                                                                                                                                  | 0.000                        | 21.120              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 10797.600                                                                          | 8,741.899                                                                                                       |
| 25/06/2013           | 25/07/2013 | 17169.240                                                                                                      | 17169.240                   | 17169.240          | 33.000                                                                                                                                  | 35.640                       | 35.640              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 9275.200                                                                           | 7,857.939                                                                                                       |
| 26/07/2013           | 25/08/2013 | 10152.120                                                                                                      | 10152.120                   | 10152.120          | 42.240                                                                                                                                  | 44.880                       | 44.880              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 5427.840                                                                           | 4,678.939                                                                                                       |
| 26/08/2013           | 24/09/2013 | 11993.520                                                                                                      | 11993.520                   | 11993.520          | 56.760                                                                                                                                  | 62.040                       | 62.040              | 2.297                                                                                                                                 | 2.297                        | 2.297               | 6246.240                                                                           | 5,682.943                                                                                                       |
| 25/09/2013           | 25/10/2013 | 18540.720                                                                                                      | 18540.720                   | 18540.720          | 73.920                                                                                                                                  | 64.680                       | 73.920              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 9584.960                                                                           | 8,881.379                                                                                                       |
| 26/10/2013           | 24/11/2013 | 17376.480                                                                                                      | 17376.480                   | 17376.480          | 80.520                                                                                                                                  | 73.920                       | 80.520              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 8780.640                                                                           | 8,514.859                                                                                                       |
| 25/11/2013           | 25/12/2013 | 21795.840                                                                                                      | 21795.840                   | 21795.840          | 43.560                                                                                                                                  | 59.400                       | 59.400              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 11114.400                                                                          | 10,621.579                                                                                                      |
| 26/12/2013           | 31/12/2013 | 4819.320                                                                                                       | 4819.320                    | 4819.320           | 1.320                                                                                                                                   | 7.128                        | 7.128               | 0.092                                                                                                                                 | 0.092                        | 0.092               | 2826.560                                                                           | 1,985.540                                                                                                       |
| <b>Subtotal 2013</b> |            |                                                                                                                |                             | <b>231945.120</b>  |                                                                                                                                         |                              | <b>605.088</b>      |                                                                                                                                       |                              | <b>7.345</b>        | <b>122631.520</b>                                                                  | <b>108,701.167</b>                                                                                              |
| 01/01/2014           | 24/01/2014 | 11274.120                                                                                                      | 11274.200                   | 11274.120          | 33.000                                                                                                                                  | 28.512                       | 33.000              | 0.369                                                                                                                                 | 0.369                        | 0.369               | 6052.640                                                                           | 5,188.111                                                                                                       |
| 25/01/2014           | 25/02/2014 | 17651.040                                                                                                      | 17651.040                   | 17651.040          | 40.920                                                                                                                                  | 36.960                       | 40.920              | 0.461                                                                                                                                 | 0.461                        | 0.461               | 9597.280                                                                           | 8,012.379                                                                                                       |

|               |            |           |           |            |         |         |         |        |        |        |            |            |
|---------------|------------|-----------|-----------|------------|---------|---------|---------|--------|--------|--------|------------|------------|
| 26/02/2014    | 25/03/2014 | 19215.240 | 19215.240 | 19215.240  | 34.320  | 40.920  | 40.920  | 0.461  | 0.461  | 0.461  | 9347.360   | 9,826.499  |
| 26/03/2014    | 25/04/2014 | 16573.920 | 16573.920 | 16573.920  | 92.400  | 68.640  | 92.400  | 0.461  | 0.461  | 0.461  | 9285.760   | 7,195.299  |
| 26/04/2014    | 26/05/2014 | 28915.920 | 28915.920 | 28915.920  | 35.640  | 54.120  | 54.120  | 0.461  | 0.461  | 0.461  | 14567.520  | 14,293.819 |
| 27/05/2014    | 24/06/2014 | 16312.560 | 16312.560 | 16312.560  | 62.040  | 46.200  | 62.040  | 0.461  | 0.461  | 0.461  | 8050.240   | 8,199.819  |
| 25/06/2014    | 26/07/2014 | 13211.880 | 13211.880 | 13211.880  | 50.160  | 71.280  | 71.280  | 0.461  | 0.461  | 0.461  | 7143.840   | 5,996.299  |
| 27/07/2014    | 26/08/2014 | 7539.840  | 7539.840  | 7539.840   | 91.080  | 75.240  | 91.080  | 0.461  | 0.461  | 0.461  | 3706.560   | 3,741.739  |
| 27/08/2014    | 25/09/2014 | 4623.960  | 4623.960  | 4623.960   | 120.120 | 122.760 | 122.760 | 0.461  | 0.461  | 0.461  | 2455.200   | 2,045.539  |
| 26/09/2014    | 26/10/2014 | 16060.440 | 16060.440 | 16060.440  | 33.000  | 44.880  | 44.880  | 0.461  | 0.461  | 0.461  | 8608.160   | 7,406.939  |
| 27/10/2014    | 24/11/2014 | 14810.400 | 14810.400 | 14810.400  | 81.840  | 75.240  | 81.840  | 0.461  | 0.461  | 0.461  | 7080.480   | 7,647.619  |
| 25/11/2014    | 25/12/2014 | 27009.840 | 27009.840 | 27009.840  | 39.600  | 52.800  | 52.800  | 0.461  | 0.461  | 0.461  | 14078.240  | 12,878.339 |
| 26/12/2014    | 31/12/2014 | 3033.360  | 3033.360  | 3033.360   | 22.440  | 11.886  | 22.440  | 0.086  | 0.086  | 0.086  | 1561.120   | 1,449.714  |
| Subtotal 2014 |            |           |           | 196232.520 |         |         | 810.480 |        |        | 5.526  | 101534.400 | 93882.113  |
| 01/01/2015    | 25/01/2015 | 15265.800 | 15265.800 | 15265.800  | 40.920  | 47.514  | 47.514  | 0.375  | 0.375  | 0.375  | 8210.400   | 7,007.511  |
| 26/01/2015    | 25/02/2015 | 13902.240 | 13902.240 | 13902.240  | 68.640  | 62.040  | 68.640  | 0.461  | 0.461  | 0.461  | 7332.160   | 6,500.979  |
| 26/02/2015    | 25/03/2015 | 18724.200 | 18724.200 | 18724.200  | 36.960  | 40.920  | 40.920  | 0.461  | 0.461  | 0.461  | 9241.760   | 9,441.059  |
| 26/03/2015    | 25/04/2015 | 26071.320 | 26071.320 | 26071.320  | 18.480  | 23.760  | 23.760  | 6.083  | 6.083  | 6.083  | 13759.680  | 12,281.797 |
| 26/04/2015    | 25/05/2015 | 21658.560 | 21658.560 | 21658.560  | 30.360  | 29.040  | 30.360  | 0.461  | 0.461  | 0.461  | 11098.560  | 10,529.179 |
| 26/05/2015    | 25/06/2015 | 18256.920 | 18256.920 | 18256.920  | 40.920  | 34.320  | 40.920  | 0.461  | 0.461  | 0.461  | 9771.520   | 8,444.019  |
| 26/06/2015    | 25/07/2015 | 10775.160 | 10775.160 | 10775.160  | 43.560  | 51.480  | 51.480  | 0.461  | 0.461  | 0.461  | 5624.960   | 5,098.259  |
| 26/07/2015    | 25/08/2015 | 8338.440  | 8338.440  | 8338.440   | 89.760  | 72.600  | 89.760  | 3.766  | 3.766  | 3.766  | 4241.600   | 4,003.314  |
| 26/08/2015    | 25/09/2015 | 7213.800  | 7213.800  | 7213.800   | 85.800  | 89.760  | 89.760  | 22.177 | 22.177 | 22.177 | 3991.680   | 3,110.183  |
| 26/09/2015    | 25/10/2015 | 15859.800 | 15859.800 | 15859.800  | 54.120  | 52.800  | 54.120  | 20.222 | 20.222 | 20.222 | 7835.520   | 7,949.938  |
| 26/10/2015    | 25/11/2015 | 17589.000 | 17589.000 | 17589.000  | 59.400  | 71.280  | 71.280  | 8.108  | 8.108  | 8.108  | 9570.880   | 7,938.732  |
| 26/11/2015    | 26/12/2015 | 13702.920 | 13702.920 | 13702.920  | 21.120  | 15.840  | 21.120  | 9.858  | 9.858  | 9.858  | 6834.080   | 6,837.862  |
| 27/12/2015    | 31/12/2015 | 2566.080  | 2566.080  | 2566.080   | 6.600   | 11.220  | 11.220  | 0.077  | 0.077  | 0.077  | 1092.960   | 1,461.823  |
| Subtotal 2015 |            |           |           | 189924.240 |         |         | 640.854 |        |        | 72.971 | 98605.760  | 90604.655  |
| 01/01/2016    | 26/01/2016 | 18068.160 | 18068.160 | 18068.160  | 54.120  | 56.100  | 56.100  | 0.384  | 0.384  | 0.384  | 9292.800   | 8,718.876  |
| 27/01/2016    | 24/02/2016 | 17657.640 | 17657.64  | 17657.640  | 40.920  | 43.560  | 43.560  | 0.461  | 0.461  | 0.461  | 9109.760   | 8,503.859  |
| 25/02/2016    | 26/03/2016 | 18757.200 | 18757.20  | 18757.200  | 29.040  | 29.040  | 29.040  | 0.461  | 0.461  | 0.461  | 10035.520  | 8,692.179  |
| 27/03/2016    | 25/04/2016 | 23687.400 | 23687.40  | 23687.400  | 31.680  | 29.040  | 31.680  | 0.461  | 0.461  | 0.461  | 12242.560  | 11,412.699 |
| 26/04/2016    | 26/05/2016 | 23040.600 | 23040.60  | 23040.600  | 30.360  | 26.400  | 30.360  | 0.461  | 0.461  | 0.461  | 11792.000  | 11,217.779 |

|                      |            |           |           |                    |         |         |                 |       |       |                |                   |                   |
|----------------------|------------|-----------|-----------|--------------------|---------|---------|-----------------|-------|-------|----------------|-------------------|-------------------|
| 27/05/2016           | 24/06/2016 | 14506.800 | 14506.80  | 14506.800          | 48.840  | 52.800  | 52.800          | 0.461 | 0.461 | 0.461          | 7935.840          | 6,517.699         |
| 25/06/2016           | 26/07/2016 | 17466.240 | 17466.24  | 17466.240          | 46.200  | 50.558  | 50.558          | 1.383 | 1.383 | 1.383          | 8986.560          | 8,427.739         |
| 27/07/2016           | 26/08/2016 | 7702.200  | 7702.20   | 7702.200           | 91.080  | 71.280  | 91.080          | 0.461 | 0.461 | 0.461          | 4341.920          | 3,268.739         |
| 27/08/2016           | 26/09/2016 | 13700.280 | 13700.28  | 13700.280          | 79.200  | 97.680  | 97.680          | 0.461 | 0.461 | 0.461          | 6825.280          | 6,776.859         |
| 27/09/2016           | 26/10/2016 | 12980.880 | 12980.88  | 12980.880          | 80.520  | 71.280  | 80.520          | 1.619 | 1.619 | 1.619          | 6696.800          | 6,201.941         |
| 27/10/2016           | 26/11/2016 | 17926.920 | 17926.92  | 17926.920          | 52.800  | 59.400  | 59.400          | 0.461 | 0.461 | 0.461          | 9238.240          | 8,628.819         |
| 27/11/2016           | 26/12/2016 | 15553.560 | 15553.56  | 15553.560          | 68.640  | 51.480  | 68.640          | 0.461 | 0.461 | 0.461          | 8377.600          | 7,106.859         |
| 27/12/2016           | 31/12/2016 | 3340.920  | 3340.920  | 3340.920           | 14.520  | 16.720  | 16.720          | 0.077 | 0.077 | 0.077          | 1344.640          | 1,979.483         |
| <b>Subtotal 2016</b> |            |           |           | <b>204388.800</b>  |         |         | <b>708.138</b>  |       |       | <b>7.612</b>   | <b>106219.520</b> | <b>97453.530</b>  |
| 01/01/2017           | 26/01/2017 | 10350.120 | 10350.120 | 10350.120          | 67.320  | 83.600  | 83.600          | 0.384 | 0.384 | 0.384          | 6691.520          | 3,574.616         |
| 27/01/2017           | 26/02/2017 | 24303.840 | 24303.840 | 24303.840          | 48.840  | 47.520  | 48.840          | 0.461 | 0.461 | 0.461          | 11281.600         | 12,972.939        |
| 27/02/2017           | 26/03/2017 | 14939.760 | 14939.760 | 14939.760          | 26.400  | 23.760  | 26.400          | 0.461 | 0.461 | 0.461          | 7978.080          | 6,934.819         |
| 27/03/2017           | 26/04/2017 | 17109.840 | 17109.840 | 17109.840          | 54.120  | 54.120  | 54.120          | 0.461 | 0.461 | 0.461          | 9074.560          | 7,980.699         |
| 27/04/2017           | 26/05/2017 | 23114.520 | 23114.520 | 23114.520          | 43.560  | 43.560  | 43.560          | 0.461 | 0.461 | 0.461          | 11932.800         | 11,137.699        |
| 27/05/2017           | 20/06/2017 | 14199.240 | 14199.240 | 14199.240          | 40.920  | 40.920  | 40.920          | 0.461 | 0.461 | 0.461          | 7606.720          | 6,551.139         |
| 21/06/2017           | 20/07/2017 | 13824.360 | 13824.360 | 13824.360          | 33.000  | 33.000  | 33.000          | 1.148 | 1.148 | 1.148          | 7307.027          | 6,483.185         |
| 21/07/2017           | 20/08/2017 | 11556.600 | 11556.600 | 11556.600          | 102.500 | 102.960 | 102.960         | 0.461 | 0.461 | 0.461          | 5781.600          | 5,671.579         |
| 21/08/2017           | 20/09/2017 | 6585.480  | 6585.480  | 6585.480           | 100.320 | 100.320 | 100.320         | 2.999 | 2.999 | 2.999          | 3375.680          | 3,106.481         |
| 21/09/2017           | 20/10/2017 | 13152.480 | 13152.480 | 13152.480          | 67.320  | 67.320  | 67.320          | 0.461 | 0.461 | 0.461          | 6716.160          | 6,368.539         |
| 21/10/2017           | 20/11/2017 | 20602.560 | 20602.560 | 20602.560          | 64.680  | 64.680  | 64.680          | 0.461 | 0.461 | 0.461          | 11200.640         | 9,336.779         |
| 21/11/2017           | 20/12/2017 | 19080.600 | 19080.600 | 19080.600          | 40.920  | 40.920  | 40.920          | 0.461 | 0.461 | 0.461          | 10537.120         | 8,502.099         |
| <b>Subtotal 2017</b> |            |           |           | <b>188819.400</b>  |         |         | <b>706.640</b>  |       |       | <b>8.680</b>   | <b>99483.507</b>  | <b>88620.573</b>  |
| <b>Total</b>         |            |           |           | <b>1075354.250</b> |         |         | <b>3647.124</b> |       |       | <b>103.600</b> | <b>563813.747</b> | <b>507789.778</b> |

The corresponding baseline emission reductions during this monitoring period is calculated as:

| Monitoring period |            | Net electricity supplied to the grid by the project ( $EG_{\text{facility}}$ ) unit: MWh | Emission Factor ( $EF_{\text{grid,CM,y}}$ ) unit: $tCO_2e/MWh$ | Baseline Emission ( $BE_y$ ) unit: $tCO_2e$ |
|-------------------|------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------|
| From              | To         |                                                                                          |                                                                |                                             |
| 28/09/2012        | 31/12/2012 | 28,527.740                                                                               | 0.9309                                                         | 26,556                                      |
| 01/01/2013        | 31/12/2013 | 108,701.167                                                                              | 0.9309                                                         | 101,189                                     |
| 01/01/2014        | 31/12/2014 | 93,882.113                                                                               | 0.9309                                                         | 87,394                                      |
| 01/01/2015        | 31/12/2015 | 90,604.655                                                                               | 0.9309                                                         | 84,343                                      |
| 01/01/2016        | 31/12/2016 | 97,453.530                                                                               | 0.9309                                                         | 90,719                                      |
| 01/01/2017        | 20/12/2017 | 88,620.573                                                                               | 0.9309                                                         | 82,496                                      |
| <b>Total</b>      |            | <b>507,789.778</b>                                                                       | <b>-</b>                                                       | <b>472,697</b>                              |

### Project emission

As statement in the registered VCS PD, for the wind power activities, the project emissions from the project is zero. Hence,  $PE_y$  during the monitoring period from the Project is zero.

### Leakages

Leakage does not need to be accounted for this project as per the monitoring methodology ACM0002 (version 12.2.0).

### Emission reduction

The emission reductions for this monitoring period were calculated as:

| Monitoring period                       | GHG emission reductions or removals ( $tCO_2e$ ) |
|-----------------------------------------|--------------------------------------------------|
| 28/09/2012-20/12/2017                   | <b>472,697</b>                                   |
| <b>Total ERs claimed (in 1910 days)</b> | <b>472,697</b>                                   |

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates in registered VCS-PD.

During the monitoring period, the actual emission reduction of 1910 days is 472,697 $tCO_2e$ , which is 0.02% lower than the estimated emission reduction of 482,382 $tCO_2e$ , which will not cause the over estimation of emission reductions. The deviation is within a reasonable range.

Therefore, CQC is able to confirm that the actual power supply and also emission reductions reported in this monitoring period are reasonable and appropriate. CQC verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent.

## 4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions.

CQC concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission reductions is reliable.

## 4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

# 5 VERIFICATION CONCLUSION

CQC has performed the verification of the emission reductions that have been reported for the project activity “QINGDAO LONGXIN WIND POWER PROJECT PHASE I” in China (VCS Project ID: 969) for the period 28-September-2012 to 20-December-2017.

The verification is based on the baseline and monitoring methodology ACM0002 (version 12.2.0), the VCS PD. The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

Our verification approach was based on the requirements as defined under the applicable VCS Standard Version 4.3 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered VCS PD ;
- the monitoring plan is as per the applied methodology;
- the monitoring complies with the monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard Version 4.3 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;

- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

In CQC's opinion the GHG emissions reductions of the "QINGDAO LONGXIN WIND POWER PROJECT PHASE I" for the period 28/09/2012-20/12/2017 are fairly stated in the monitoring report(version 02 dated 30/08/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 12.2.0) and the monitoring plan contained in the registered VCS PD.

CQC can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CQC confirms the following statement:

Reporting period: 28/09/2012-20/12/2017

Verified GHG emission reductions and removals in the above verification period:

| Year                  | Baseline emissions or removals(tCO <sub>2</sub> e) | Project emissions or removals (tCO <sub>2</sub> e) | Leakage emissions (tCO <sub>2</sub> e) | Net GHG emission reductions or removals(tCO <sub>2</sub> e) |
|-----------------------|----------------------------------------------------|----------------------------------------------------|----------------------------------------|-------------------------------------------------------------|
| 28/09/2012-31/12/2012 | 26,556                                             | 0                                                  | 0                                      | 26,556                                                      |
| 01/01/2013-31/12/2013 | 101,189                                            | 0                                                  | 0                                      | 101,189                                                     |
| 01/01/2014–31/12/2014 | 87,394                                             | 0                                                  | 0                                      | 87,394                                                      |
| 01/01/2015–31/12/2015 | 84,343                                             | 0                                                  | 0                                      | 84,343                                                      |
| 01/01/2016–31/12/2016 | 90,719                                             | 0                                                  | 0                                      | 90,719                                                      |
| 01/01/2017–20/12/2017 | 82,496                                             | 0                                                  | 0                                      | 82,496                                                      |
| <b>Total</b>          | <b>472,697</b>                                     | <b>0</b>                                           | <b>0</b>                               | <b>472,697</b>                                              |



# APPENDIX A: REFERENCE

Documentation used to verify the information provided by the project proponents

|      |                                                                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /1/  | Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.: VER/VCU Monitoring Report for Qingdao Longxin Wind Power Project Phase I, version 02dated 30/08/2022                           |
| /2/  | Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.: Emission reduction calculation spreadsheet for Qingdao Longxin Wind Power Project Phase I, version 2.0 dated 06/09/2022        |
| /3/  | China Resources Wind Power (Qingdao) Co., Ltd.: Purchase Agreement for Qingdao Longxin Wind Power Project Phase I                                                                                     |
| /4/  | China Resources Wind Power (Qingdao) Co., Ltd.: Diagram of power connection system                                                                                                                    |
| /5/  | China Resources Wind Power (Qingdao) Co., Ltd.: VER monitoring manual and management procedure.                                                                                                       |
| /6/  | China Resources Wind Power (Qingdao) Co., Ltd.: Records of training for on-site staff: Records of training for on-site staff.                                                                         |
| /7/  | China Resources Wind Power (Qingdao) Co., Ltd.: Daily operation and maintenance records.                                                                                                              |
| /8/  | China Resources Wind Power (Qingdao) Co., Ltd.: Monthly and daily reading records Qingdao Longxin Wind Power Project Phase I from 28/09/2012-20/12/2017                                               |
| /9/  | State Grid Shandong Power Company: Monthly electricity receipts from 28/09/2012-20/12/2017 & Confirmation letter for electricity readings the period 28/09/2012-20/12/2017 issued by the grid company |
| /10/ | Comment records                                                                                                                                                                                       |
| /11/ | Contact information of PP posted on the bulletin board of village committee                                                                                                                           |
| /12/ | Yearly questionnaires for local stakeholders                                                                                                                                                          |
| /13/ | China Resources Wind Power (Qingdao) Co.: Registered CDM PDD                                                                                                                                          |

|      |                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /14/ | State Grid Shandong Electric Power Metering Center/ Shandong Institute of Metrology : Calibration certificates for meters M1,M2 ,M3 and M4 covering this monitoring period              |
| /15/ | Business license of China Resources Wind Power (Qingdao) Co., Ltd.                                                                                                                      |
| /16/ | State Economic and Trade Commission: Technical administrative code of electric energy metering (DL/T 448-2000, DL/T 448-2016)                                                           |
| /17/ | UNFCCC, <a href="http://cdm.unfccc.int">http://cdm.unfccc.int</a>                                                                                                                       |
| /18/ | Gold Standard, <a href="http://www.goldstandard.org/">http://www.goldstandard.org/</a>                                                                                                  |
| /19/ | China CCER Exchange Info-platform, <a href="http://cdm.ccchina.org.cn/ccer.aspx">http://cdm.ccchina.org.cn/ccer.aspx</a>                                                                |
| /20/ | Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020                                                    |
| /21/ | Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China                                                                               |
| /22/ | China Resources Wind Power (Qingdao) Co., Ltd.: Staff roster and payroll in 2017                                                                                                        |
| /23/ | China Resources Wind Power (Qingdao) Co., Ltd.: the Explanation of the employment offered by the project.                                                                               |
| /24/ | National poverty standard in 2008 and 2016<br><a href="https://www.gpbctv.com/jrtd/202104/159842.html?btwaf=54204711">https://www.gpbctv.com/jrtd/202104/159842.html?btwaf=54204711</a> |
| /25/ | TÜV Rheinland (China) Ltd. : VCS Validation Report for the project                                                                                                                      |
| /26/ | VER/VCU Monitoring Report for the project from 08/11/2011- 27/09/2012                                                                                                                   |
| /27/ | TÜV Rheinland (China) Ltd.: VCS Verification Report for the project from 08/11/2011- 27/09/2012                                                                                         |
| /28/ | China Resources Wind Power (Qingdao) Co., Ltd.: Nameplates of the equipment                                                                                                             |
| /29/ | Stakeholder Consultation Opinion Response policy of Qingdao Longxin Wind Power Project Phase I                                                                                          |

|      |                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------|
| /30/ | Feasibility Study Report of Qingdao Longxin Wind Power Project Phase I                                                               |
| /31/ | Photo of grievance book to collect comments about project implementation from local stakeholders                                     |
| /32/ | China Resources Wind Power (Qingdao) Co., Ltd.: Construction completion report of the Project                                        |
| /33/ | China Resources Wind Power (Qingdao) Co., Ltd.: Wind Turbines Purchase Agreement, dated 31/03/2007: Wind Turbines Purchase Agreement |

## Methodologies, tools and other guidance

|      |                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------|
| /37/ | Verified Carbon Standard: VCS Standard, version 4.3                                                                |
| /38/ | Verified Carbon Standard: VCS Program Guide, version 4.1                                                           |
| /39/ | Verified Carbon Standard: VCS Sectoral Scopes<br><a href="http://v-c-s.org/node/448">http://v-c-s.org/node/448</a> |
| /40/ | Verified Carbon Standard: Registration and Issuance Process, version 4.1                                           |
| /41/ | UNFCCC EB: Approved methodology, ACM0002 , version12.2.0                                                           |

# APPENDIX B: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

| CAR ID | Corrective Action Request                                                                                                                                                                                   | Response by Project Proponent                                                                                                                                                                                                                                                                                                                                                                           | Verification Team Assessment                                                                            |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| CAR 01 | The monitoring period in the MR(from 28/09/2012 to 07/11/2021, which is more than six consecutive years) does not meet the requirements of rotation of validation/verification bodies in VCS-Standard_v4.3. | The monitoring period has revised from 28/09/2012-07/11/2021 to 28/09/2012-20/12/2017.                                                                                                                                                                                                                                                                                                                  | OK.<br><br>It has been revised and verified by CQC by checking updated MR. Therefore, CAR 01 is closed. |
| CAR 02 | The information of the monitoring meters installed during the monitoring period is not provided in the MR and supporting documents.                                                                         | The information of the monitoring meters during this monitoring period have been added in the section 4.1 of the revised MR (V02, 30/08/2022). The main information are including the type, accuracy, serial number, calibration date, validity period and name of the calibration organization, and the meters' photo and the calibration reports covering this monitoring period have been submitted. | OK.<br><br>It has been revised and verified by CQC by checking updated MR. Therefore, CAR 01 is closed. |

Table 2: Clarification Requests

| CL ID | Clarification Request | Response by Project Proponent | Verification Team Assessment |
|-------|-----------------------|-------------------------------|------------------------------|
|-------|-----------------------|-------------------------------|------------------------------|

|    |    |    |    |
|----|----|----|----|
| NA | NA | NA | NA |
|----|----|----|----|

Table 3: Forward Action Requests

| <b>FAR ID</b> | <b>Forward Action Request</b> | <b>Response by Project Proponent</b> | <b>Verification Team Assessment</b> |
|---------------|-------------------------------|--------------------------------------|-------------------------------------|
| NA            | NA                            | NA                                   | NA                                  |